

Junior Engineer Technical Task

Welcome to the initial step of our application process! This task is designed to assess your coding skills, problem-solving capabilities, and creativity. If you pass this step, please note that an extended version of the task will be discussed in the next step, which is the interview, so we strongly encourage you to complete it independently.

Task Overview:

Write a simple executable program that selects a desktop wallpaper from a given set of images. The selected wallpaper should approximately represent the location of the sun at the time of execution, based on the given geographical location. **This task should not take more than 4 hours to complete.**

Details:

- You can use any generic programming language of your choice.
- You can use the programming language's standard library, any HTTP library, and any publicly available APIs. External libraries are not allowed.
- You can use any resource available on the Internet and LLMs to help you in writing the program.
- The location is given to the program as latitude and longitude in the first and second arguments, respectively.
- The set of images to be used is available in the attached zip archive.
- The program is expected to output the image file name in the `stdout`.
- If the sun is below the horizon, always output `night.png` as the file name.
- The program is expected to run in a Linux Bash environment.
- Assume that the system time where the program is executed is correct.
- Assume the given latitude and longitude are within areas where the sun sets and rises every day.

Submission Format: Submit your solution via this [1] form as a ZIP archive containing all necessary files. Please name your ZIP archive in the following format: 'FirstName_LastName_EmailAddress_Date.zip'. This is crucial for tracking your submission efficiently. Failure to follow this format may result in your application not being processed. [1]: <https://forms.gle/WU9MbpbiixN38xTq5>

Example Outputs:

Below is the expected output if the program is executed at 2024-06-01 07:30:00 in the local timezone of the given location, with the corresponding input arguments.

Example #	1st Argument	2nd Argument	Output
1	31.9544	35.9106	morning.png
2	-37.8136	144.9631	sunrise.png
3	-51.63092	-69.2247	night.png